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Digital twin-enabled prefabrication supply chain for smart construction and carbon emissions evaluation in building projects

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ABSTRACT

Digital twin (DT) provides effective pathways to solve issues in the construction industry, particularly smart construction and carbon emissions in prefabrication. Past DT research explored facility management and fault detection, highlighting a knowledge gap on the use of DT for smart construction and emissions monitoring in prefabrication supply chain (PSC). Therefore, the aim of this study is to present a holistic view of DT applications in PSC by exploring real-time smart construction and carbon emissions monitoring. A mixed-method review was adopted in two-steps involving scientometric and qualitative analysis in this study. Findings from the scientometric analysis revealed high interest in research themes such as emissions and energy control, artificial intelligence-based decision-making and blockchain integration in DT for prefabrication. Furthermore, the findings from the qualitative analysis demonstrated how smart technologies such as radio frequency identification (RFID), global positioning systems (GPS), laser scanners and sensors have been employed at the production, transportation, and on-site assembly stages of PSC for buildings. For real-time carbon emissions monitoring in DT, this study revealed various smart technologies and their corresponding information requirements for materials/components, machinery, and processes at each stage of the PSC. Five future research directions were provided on effective ways to advance DT in PSC for intelligent building processes and monitor emissions. Therefore, this study not only shows smart technologies suitable for DT in PSC, but also contributes to knowledge on using DT to monitor real-time carbon emissions in PSC for buildings. This study would aid researchers and practitioners with systemic approaches to employ when applying DT in PSC.

1. Introduction

The technological advancements in the architecture, engineering, and construction (AEC) industry have been rapid in recent years [1,2]. Most of these technologies aim to improve the performance and management of processes and buildings by enhancing intelligence and reducing emissions [3]. Among these technological advancements, the concept of digital twin (DT) offers an overarching sphere to monitor, manage and evaluate technological systems in AEC [4]. Even though DT is a relatively new subject within AEC, it

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has great capabilities in solving problems plaguing the built environment. Areas where some researchers have explored or applied DT include facility management, geometric quality assessment and engineering processes [5–7]. However, DT applications for smart construction have received limited attention in the literature, despite the increase in smart construction developments. On top of that, the potential of DT for carbon emissions monitoring seems missing in these emerging research explorations. Specifically, DT's potential to evaluate the embodied emissions associated with prefabricated construction is lacking in the evolving construction body of knowledge. The current drive to advance prefabrication with smart technologies and reduce carbon emissions [8], puts the spotlight on the contribution of DT towards these advancements. Past studies have indicated that incorporating smart construction and carbon emissions into prefabrication promises to transform construction and building delivery [9,10]. Hence, there is high interest from researchers, practitioners, and governments in making prefabricated construction activities more environmentally sustainable.

Prefabrication has gained popularity as an innovative approach to advance and decarbonize construction processes and activities [11,12]. Prefabrication has the advantages of reducing environmental impacts and promoting smart technologies in construction processes [13,14]. In contrast, the prefabrication supply chain (PSC) is responsible for embodied emissions and the application of smart technologies in the AEC industry [15]. However, in the literature, there is a paucity of knowledge on the intersection of smart technologies and carbon emissions in prefabrication. More importantly, the possibilities for real-time monitoring of carbon emissions in the prefabrication supply chain (PSC) have not attracted much attention. This has resulted in having limited knowledge of how smart construction has been employed to advance prefabrication and how carbon emissions in the various processes of PSC would be assessed in real-time. Applying DT to smart construction and carbon emissions in PSC is important to answer critical questions pertaining to: (i) smart technology applications in PSC, and (ii) essential areas for real-time carbon emissions evaluation in PSC.

The emerging literature on DT explores smart construction and carbon reduction in PSC separately, and this has tendencies to obscure the ameliorating opportunities that DT offer in PSC. Although, there is a scholarly debate on the real contributions of prefabrication to carbon reduction targets [16], most of the previous studies paid little attention to DT concerning prefabrication. Rather, these studies focused DT research on facility operations and maintenance processes. Therefore, this study aims to present a holistic view of DT applications in PSC for smart construction and carbon emissions. With a knowledge gap on ways in which DT supports smart construction and carbon emissions evaluation, this study offers systematic approaches to facilitate real-time smart construction and carbon emissions monitoring in PSC. Furthermore, by exploring the possibilities of digitally twinning PSC for carbon emissions, this study adds value to the body of knowledge on continuous evaluation of carbon emissions in real-time as compared to one-off calculations conducted at the design stage of buildings. This is of great necessity to researchers and practitioners for real-time continuous evaluation of emissions in buildings. To achieve the study's aim, two main objectives were formulated: (1) to explore the knowledge landscape of DT regarding smart construction and carbon emissions in prefabrication; and (2) to assess DT applications at various PSC stages regarding smart construction and carbon emissions. The reminder of this paper is organized as background and related work, followed by research methodology and results for smart construction and carbon emissions. Subsequent sections include discussions and future research directions, implications and conclusions.

2. Background and related work

2.1. DT in construction

The history of DT in construction is relatively brief compared to its emergence in the manufacturing and aerospace industries [17,18]. However, this evolving technology/concept is developing at a rapid pace and seems to be a critical agenda for many construction industries. This is because of the benefits it provides for real-time status and management of processes and systems (Royal Institution of Chartered Surveyors (RICS), 2022). Due to its evolution in literature, there is no universal agreement on the definition of DT. Hence, there are several definitions available from many domains. [19] envisioned product lifecycle management in a digital platform, and [20] defined DT as a set of virtual information constructs describing an actual physical product. [21] defined DT as a model representing real machine operation for simulation. [22] viewed DT as a digital counterpart of a physical product, and [23] defined DT as a digital representation of a real-world object/product in a lifecycle. These definitions forge the idea of integrating the physical world into digital environments. Nonetheless, few studies have highlighted that some of these definitions lack clarity on the constituents of DT [24]; particularly in the AEC industry. Therefore, further definitions of DT have been provided to capture the functions and characteristics of DT apart from the physical world and digital/virtual representation. Accordingly [25] provided three key elements that define DT – physical object/asset, digital/virtual counterpart and a connection between the two. Usually, this connection involves the synchronization of data and information in a bi-directional mode between the digital model and physical reality. According to Ref. [26]; the Center for Digital Built Britain (CDBB) defined DT as a realistic digital representation of assets, processes, or systems in the built environment. With this definition, emphasis is placed on closing the loop between digital models of construction and their physical twin objects with real-time information updates. Furthermore, the construction digital twin approach views DT as virtual representations of real-world systems and processes that are integrated at a specific frequency and accuracy [26,27]. By exploring the constituents and characteristics of DT, the main elements of DT consist of: (i) real-world or physical object/process, (ii) virtual or digital representation, and (iii) connection/synchronization using sensing technologies for data, real-time monitoring and the possibility of intervention based on simulation and artificial intelligence (AI).

In AEC, DT can be created in three levels; a component, a system, or a system of systems assembly of buildings and processes in an interconnected ecosystem to reflect the real occurrence of events/activities [27]. Changes in the physical environment must be replicated in the virtual twin model [28]. With the aid of sensing technologies, DT leverages two key things – (i) geometry (form) or assembly relationship state of practice, and (ii) connectivity for real-time simulations and AI-enabled analysis in decision-making.

2.2. The BIM dilemma for DT

There are debates on the use of building information modelling (BIM) models to represent DT in construction literature. BIM – a digital platform for object modelling, is increasingly becoming a common data environment that fills the need for digital prototyping in design and construction [6,26]. Therefore, many practitioners view BIM as a key element to the DT evolution in AEC. As such, some studies have perceived '*as-built*' BIM for facility and asset management as DT [29]. In contrast, [30] highlighted that digital models of assets that do not automatically interact with physical assets are digital shadows. The CDBB argues that DT goes beyond BIM creating a virtual model of a physical object [31]. DT considers continuous connectivity and real-time monitoring throughout the project lifecycle, which current *'as-built'* BIM models fall short of. Other studies differentiated BIM models from DT by highlighting that the presence of a real-world physical object is not mandatory in BIM models, and when a physical object is present (e.g. *as-built* BIM), it lacks connections with the physical part in real-time [17]. Considering that BIM has gained prominence as a common data environment for as-built facilities, its growth to become DT is to have automatic connectivity with the physical object/process in real-time throughout the project lifecycle. Thus, BIM models can be perceived as DT when virtual models represent their as-built assets/processes, and synchronous information exchange occurs between the physical and virtual twin in real-time.

2.3. Prefabrication supply chain

Prefabrication has become an innovative technology in many countries to reduce carbon emissions and mitigate climate change regarding construction activities (Oti-Sarpong et al., 2022). The concept of prefabrication is to produce building components and modules in a factory-controlled environment and assemble them at their destination on-site [32,33]. There are many variations in the prefabrication concept such as off-site construction, modular construction, design for manufacture and assembly (DfMA) and industrialized construction [34]. However, these variations do not deviate from the core concept of prefabrication [11]. Thus, shifting most of the construction activities off-site to a factor-controlled environment. Amidst the optimism surrounding advanced technologies in prefabrication, the extent to which embodied emissions are reduced at various stages of the PSC is still unclear and vague in research [15]. This presents opportunities for DT to monitor real-time emissions with smart technologies. PSC typically involves activities such as the manufacturing of building materials, transportation, factory production processes and on-site installation processes. Although each of these activities may have its respective sub-activities, they can be aggregated into three main stages of PSC; (i) production, (ii) transportation, and (iii) on-site installation for DT to explore smart construction and evaluate emissions in real-time. The prefabrication production stage focuses on building materials and factory-related processes for prefabricated components [35]. Further, the transportation stage focuses on transport-related activities and the on-site installation stage addresses assembling processes for prefabricated components [36]. For these three stages, smart construction technologies and the associated emissions would be examined.

2.4. Embodied carbon emissions in prefabricated buildings

Embodied carbon emissions (ECE) are emissions associated with producing materials and products for construction and buildings [37]. It also includes emissions from processes and activities used to transform these materials and products for buildings [38]. ECE in construction can be cumulatively stratified from material extraction to the end-of-construction [15]. In that regard, ECE is derived from the energy consumed in extracting and transporting building materials, in addition to the emissions from processes and activities of prefabrication [25,39]. This provides an indication of the sources of ECE in prefabrication, which can hence be summarized in terms of materials and processes involved. Within each stage of the PSC, these two sources of ECE are linked to key emission indicators involving resource and energy consumption, equipment used, production chain and construction activities. The respective sources of ECE for the stages of PSC are presented in Fig. 1.

To explore the potential of DT in this study, the key emission sources of ECE in the main stages of PSC will be explored in tandem with the main constituents required for DT. By this, the possibilities of applying technologies at each stage of the PSC for automatic real-time carbon emission evaluation in DT would be explored.

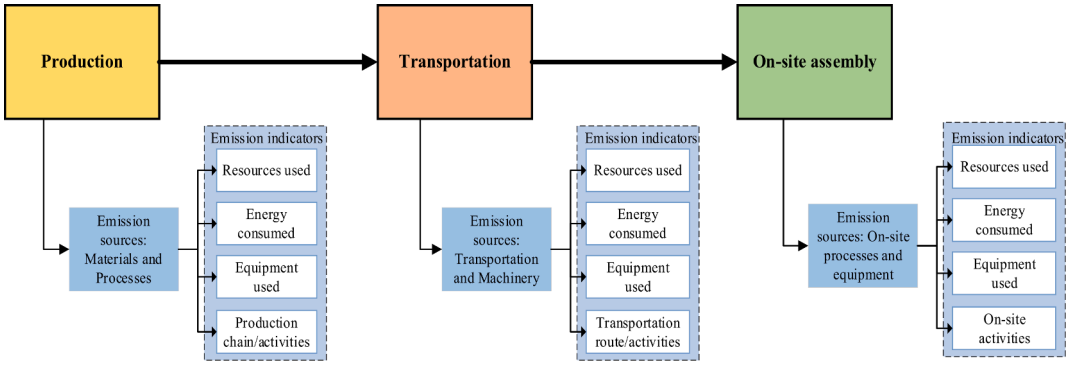


Fig. 1. Stages of prefabrication supply chain and carbon emission sources.

3. Research methodology

This study adopted the mixed-review approach in two-steps to investigate DT research regarding smart construction and emissions in prefabrication. In the first step, scientometric analysis was used to explore and map the knowledge landscape of DT research in prefabrication [40,41]. This creates an overview of research areas that are of interest in the literature using science mapping techniques [42]. The second step involves qualitative analysis by critically reviewing the application of DT for smart construction and carbon emissions at the main stages of PSC. Regarding this step, studies that have used or applied DT technologies to explore smart construction and assess carbon emissions at any stage of the PSC were extracted from the data collected in the first step and considered for critical examination.

3.1. Step 1: scientometric analysis

Due to its science mapping capabilities based on text mining, the scientometric approach focuses on identifying and visualizing patterns in knowledge domains [43]. The scientific database for extracting data in literature was Scopus. Scopus was selected because it has the largest index of academic literature and produces reputable results [44]. The literature search strategy began with the formulation of a list of keywords. The main components of the list of keywords were related to terms used for DT and prefabrication. Since different terminologies have been used interchangeably to represent these main components, a pilot review of related studies was used to select suitable keywords (e.g. Refs. [6,17,45]. Boolean operators (“AND” and “OR”) were used as loops to combine the strings of keywords [5]. To improve the comprehensiveness and complement the potential of BIM towards DT, as argued by some researchers as a pragmatic approach in AEC, additional sets of keywords were created using the constituents of DT as a requirement for the case of BIM. Therefore, terms such as “as-built BIM” and “real-time connection” were used to depict the physical twin and connectivity aspects of a DT if BIM is to be considered for DT in AEC. To that effect, the main components of the additional keyword string were related to terms for as-built BIM, real-time connection and prefabrication. The lists of keywords for the search are provided in Table 1.

3.1.1. Data collection and data cleaning

Independent searches were conducted using the two sets of keywords (Table 1) in Scopus. The inclusion criteria were peer-review journal articles [46]. These criteria were selected to improve reliable findings concerning smart construction and emissions in AEC. The initial search results from the two sets of keywords were filtered for English articles and there was no limitation on the publication year. From this process, the combined search using DT and BIM-related keywords identified a total of 136 articles. Since some articles may appear in both searches, additional filtering was conducted to remove duplicate articles in the search results. The results from these refinement processes retained 107 articles. In this study, VOSviewer was selected as the science mapping tools to analyze and visualize data obtained from the Scopus database. VOSviewer uses text mining techniques and algorithms to create networks in scientometric analysis [47]. Using this tool, the scientometric analysis explored networks and clusters pertaining to areas of research interest and global research collaborations (Fig. 2).

3.1.2. Characteristics of documents extracted

The publication trend for DT concerning PSC as presented in Fig. 3, generally depicts a sharp increase in research interest from 2018 onwards. Since DT is an emerging concept in AEC, the proliferation of smart technologies and recent calls to reduce carbon emissions may have contributed to this trend. This development indicates that not only is DT for PSC emissions attracting research attention, but it also reveals a timely intervention on the potential of DT for carbon emissions assessments in AEC.

3.1.3. Knowledge areas of research interest

Keywords reflect the essential themes explored in a topic and aid in the detection of key research areas [48]. Hence, analyzing the networks of keywords shows the relationships between topics in a knowledge domain [40]. In this research, the co-occurrence of keywords was analyzed and the network map was visualized using VOSviewer. The proximity of nodes and the size of the links depict the research interest between two or more nodes (topics) in the network [47]. The network map in Fig. 4 was developed by considering the minimum number of co-occurrences of keywords to be three, and generic keywords relating to research methods such as ‘survey’ were removed. As a result, 66 keywords were retained as they met or exceeded the criteria. Table 2 presents data from the keywords network.

As illustrated with the overlaps in Fig. 4, nodes sharing a high number of links in the network intrinsically move towards the central position. On this basis, topics such as digital twin and internet of things (IoT) have been linked the most in scholarly discussions.

Table 1
Two sets of keywords for the literature search.

Keywords for DT concerning PSC emissions	BIM related keywords towards DT for PSC emissions
"digital twin*" OR "digital replica" OR "virtual twin" OR "virtual counterpart" OR "virtual assembly twin" OR "cyber-physical systems" OR "internet of things" OR "iot" OR "smart sensors" OR "real-time monitoring" OR "real-time sensors" OR "real-time data" OR "real-time data exchange" AND "prefabricat*" OR "industrialized construction" OR "prefabricated build*" OR "prefabricated construction" OR "modular construction" OR "design for manufacture assembly" OR "dfma" OR "off-site construction" OR "modular build*" OR "modular integrated construction" OR "off-site production"	"as-built bim" OR "Building information modelling" OR "bim" AND "cyber-physical systems" OR "internet of things" OR "iot" OR "smart sensors" OR "real-time monitoring" OR "real-time sensors" OR "real-time data" OR "real-time data exchange" AND prefabricat*" OR "industrialized construction" OR "prefabricated build*" OR "prefabricated construction" OR "modular construction" OR "design for manufacture assembly" OR "dfma" OR "off-site construction" OR "modular build*" OR "modular integrated construction" OR "off-site production"

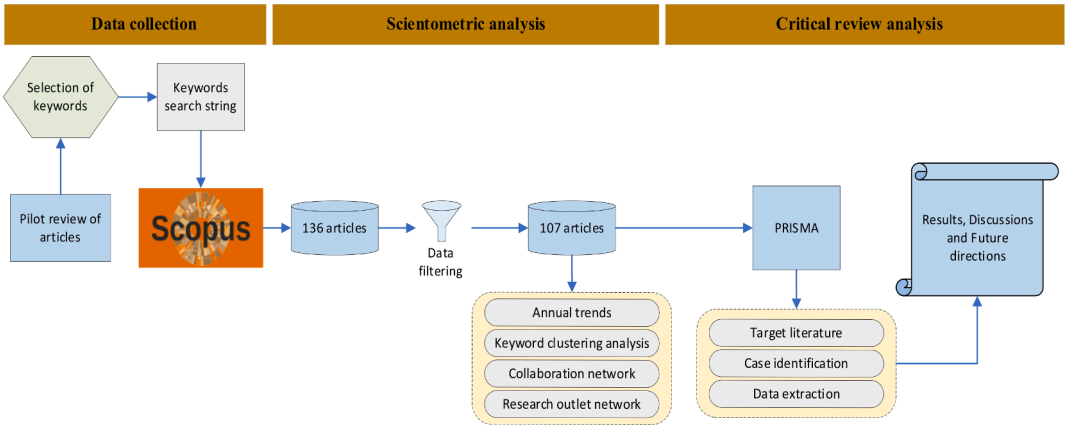


Fig. 2. Overview of research methodology.

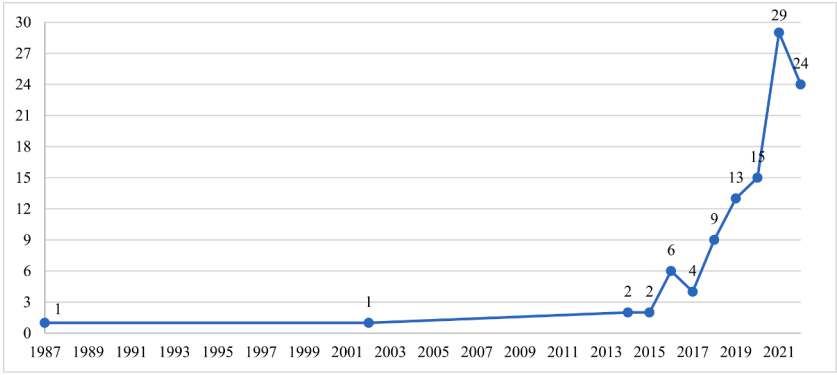


Fig. 3. Publication trend and distribution outputs.

Hence, they have proximity to each other in the network. Among these topics, IoT has the largest node size, and DT shares the strongest link strength with modular construction. While this occurrence supports the emerging interest in DT, it further consolidates the interest of researchers in promoting IoT prefabricated construction. From Fig. 4, other topics, though not at the center but nearer to the focal nodes indicate research streams evolving from DT research. These research streams include lifecycle, energy utilization, automation, decision support systems, data acquisition and Industry 4.0. Inadvertently, scholarly discussions on automating support systems to enhance smart construction and lifecycle evaluations are the underpinning drive in DT for prefabrication.

Further analysis of the network shows that topics frequently co-occurring in research studies gravitate towards each other or vice versa based on the strength of the link [49]. Hence, two close topics such as 'emission control' and 'carbon emissions' indicate that both topics have been studied frequently and therefore have a strong link between them. On the contrary, the long distance between carbon-related topics and DT or modular construction or prefabricated construction demonstrates that carbon emissions research theme is yet to gain popularity when applying DT in modular construction/prefabricated construction. Besides topics such as greenhouse gases and energy conservation being distant from DT and IoT, there exist weak links between carbon emissions research themes and DT. This reveals a relevant gap in knowledge considering the trajectory of current research for DT and prefabrication. Based on the link strength between topics, clusters were formed and colour-coded, as shown Fig. 4 and Table 2. Subsequently, five clusters were derived and named as:

- Emission and energy control in DT (Green cluster) – This cluster focuses on emissions and the associated data acquisition method for prefabrication. One major source of carbon emissions is energy. Hence, topics such as energy utilization, energy conservation, emission control and greenhouse gases have been key areas in exploring carbon emissions or GHG for prefabricated buildings and components. To evaluate energy utilized and its associated emissions, smart technologies such as radio frequency identification (RFID) and long range (LoRa) tools have been explored in tracking systems for information acquisition on prefabricated construction processes [36]. Furthermore, the potential of 3D printing by means of robotic manufacturing has attracted research interest in prefabrication.
- AI-based decision-making in DT (Red cluster) – This cluster uncovers the use of artificial intelligence to support decision-making for sustainable construction processes. Of these, modular construction processes and site assembly have been crucial in the knowledge domain. Primarily, DT is used to inform decision-making. Hence intelligence is needed to make effective decisions in real-time concerning PSC emissions. The integration of AI into DT ensures that predictive analytics are employed in determining possible scenarios that would result in low emissions for real-time decision-making. An integration of this type

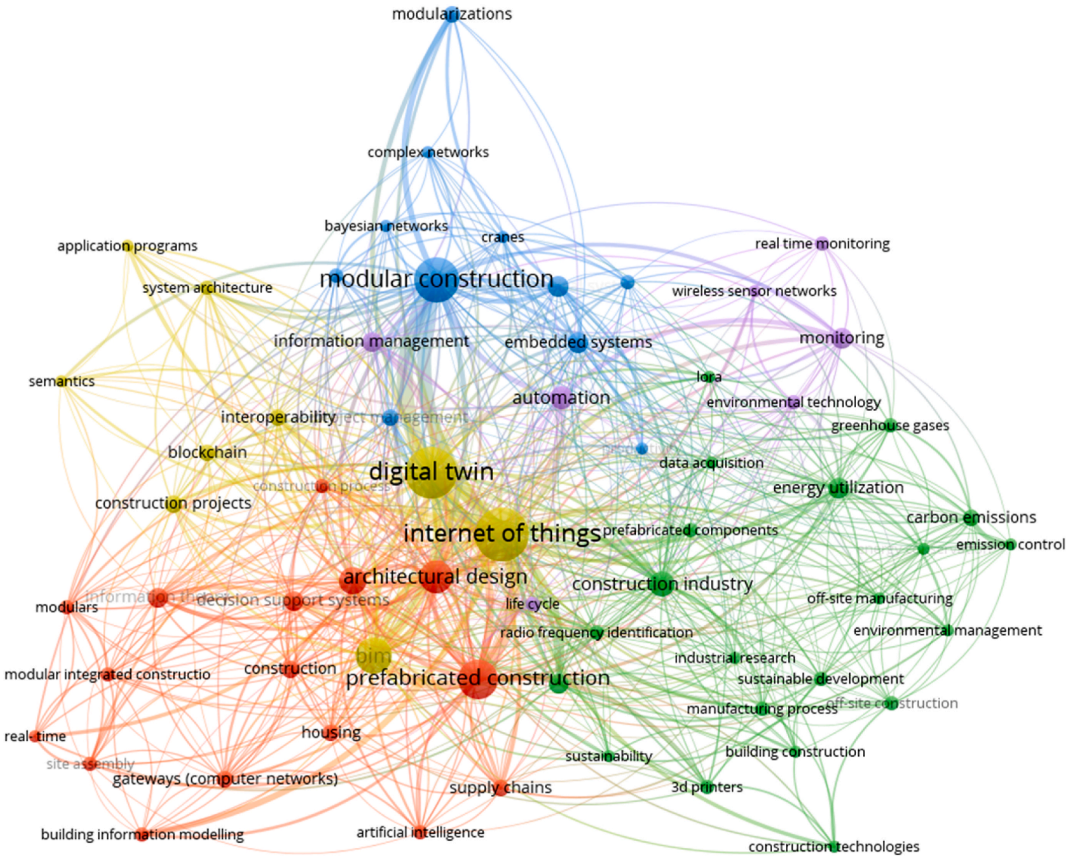


Fig. 4. Areas of interest in DT research for prefabrication.

is crucial for environmental management as carbon emissions are not only captured in DT in real-time but are also evaluated to achieve the lowest emission possible in PSC.

- Automatic monitoring systems (Purple cluster) – Monitoring emissions from PSC processes in real-time is vital if DT is to be applied. As such, automating the monitoring process for easy information retrieval and management has attracted techniques that involve wireless sensors for real-time monitoring. Various sensor devices can be used to monitor different processes and activities in PSC for emission monitoring. Typical sensors adopted in current research include global positioning systems (GPS) and pressure sensors, and they are usually integrated into a monitoring system for the automatic evaluation of emissions in prefabrication processes. More importantly, the lifecycle perspective is being integrated into the environmental technologies used for emissions monitoring.
- Visualization and management of modular construction (Blue cluster) – Efforts to advance modular construction implementation have culminated in the application of smart technologies capable of visualizing modular construction for its management. In this regard, embedded systems and data visualization have been investigated for productivity, risk assessment and equipment management. In addition, the use of virtual reality (VR) is being explored to enhance the visualization of emissions in real-time for decision-making.
- DT-and-Blockchain integration (Yellow cluster) – This cluster focuses on the development of DT with intelligent applications (Blockchain). Within this cluster, BIM is gaining traction with close ties to both DT and IoT. System architectures used in IoT and cloud computing form the foundational system for DT among complex networks of prefabrication processes and stakeholders. Hence, information signals have to be secured and transparent. With multiple processes involving different stakeholders in PSC, blockchain would facilitate trust in real-time information sharing from smart technologies from PSC collaboration networks. In this regard, blockchain enhances value in data communication among DT systems. The added value of DT-Blockchain integration is having traceable process-related information in large prefabrication teams to facilitate efficiency and sustainability.

3.1.4. Collaboration network of countries for DT in PSC emissions

The collaboration network identifies countries with an active interest in DT for PSC emissions purposes [43]. Co-authorship analysis was conducted on the data, the resultant network is presented in Fig. 5. Fig. 5 shows that China and the UK are the top two countries contributing to the utilization of DT for PSC purposes. Other significant contributors include the US, Austria, Hong Kong and Australia. Although these countries have their respective carbon emission reduction targets, more interest is needed in monitoring real-time emissions as a crucial agenda to address the decarbonization and carbon neutrality goals by 2050. For instance, with the in-

Table 2
Network data of topics in DT research for PSC emissions.

Label	Cluster	Weight (Links)	Weight (Total link strength)	Weight (Occurrences)
internet of things	4	58	49	49
digital twin	4	56	43	45
modular construction	3	45	32	36
prefabricated construction	2	53	26	27
bim	4	46	23	23
architectural design	2	46	19	19
decision making	2	36	14	14
construction industry	1	41	12	12
automation	5	25	10	10
cyber physical system	3	26	8	8
embedded systems	3	33	8	8
energy utilization	1	31	8	8
monitoring	5	24	8	8
decision support systems	2	31	7	7
industry 4.0	1	21	7	7
information management	5	27	7	7
information theory	2	27	7	7
construction projects	4	23	6	6
blockchain	4	23	5	5
carbon emissions	1	23	5	5
construction	2	22	5	6
gateways (computer networks)	2	21	5	5
housing	2	22	5	5
interoperability	4	24	5	5
modularizations	3	8	5	5
project management	3	22	5	5
supply chains	6	21	5	5
3D printers	1	16	4	4
artificial intelligence	2	18	4	4
building information modelling (bim)	2	14	4	4
construction process	2	17	4	4
data visualization	3	21	4	4
environmental technology	5	21	4	4
greenhouse gases	1	20	4	4
life cycle	5	22	4	4
manufacturing process	1	22	4	4
modular integrated construction	2	21	4	4
modulars	2	22	4	4
prefabricated components	1	24	4	4
radio frequency identification (rfid)	1	24	4	4
risk assessment	3	18	4	4
site assembly	2	17	4	4
sustainable development	1	21	4	4
system architecture	4	15	4	4
application programs	4	12	3	3
bayesian networks	3	13	3	3
building construction	1	14	3	3
complex networks	3	16	3	3
construction technologies	1	9	3	3
cranes	3	16	3	3
data acquisition	1	18	3	3
emission control	1	15	3	3
energy conservation	1	15	3	3
environmental management	1	14	3	3
industrial research	1	23	3	3
lora	1	15	3	3
off-site construction	1	20	3	4
off-site manufacturing	1	20	3	3
productivity	3	15	3	3
real- time	2	8	3	3
semantics	4	14	3	3
sustainability	1	14	3	3
wireless sensor networks	5	10	3	3
Real- time monitoring	5	13	2	4

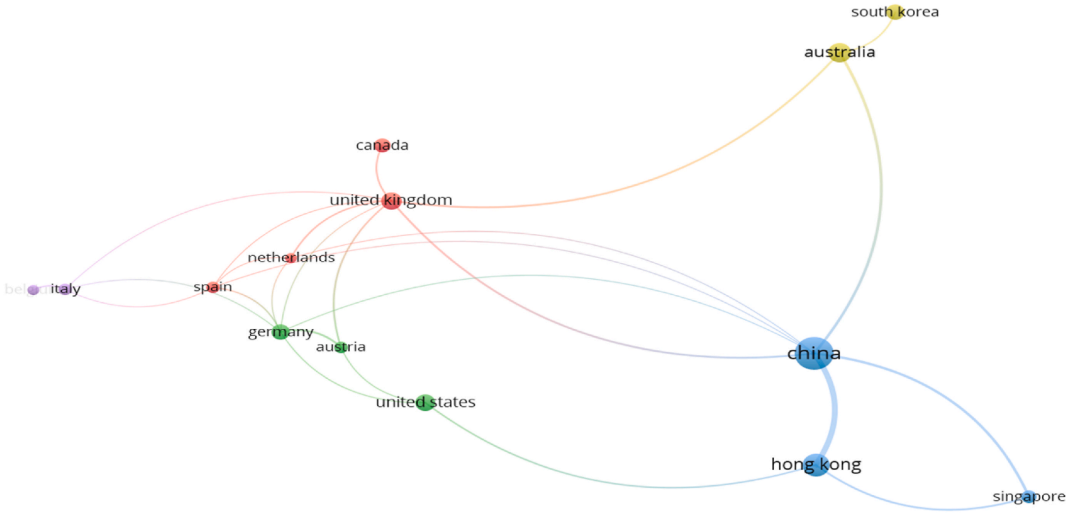


Fig. 5. Collaboration network of active countries in developing DT research for PSC emissions.

creased use of prefabricated components for housing delivery in Hong Kong and the US [15,50], improving efforts to evaluate emissions from prefabrication activities in real-time affords the opportunity to objectify the effectiveness of decarbonization or mitigation interventions.

3.1.5. Smart construction technologies adopted for DT in PSC

From Fig. 4., there are some technologies that have gained prominence in enhancing smart construction regarding DT applications for PSC. Smart technologies confer awareness and interactive abilities to objects or processes in prefabrication [51]. Such technologies include RFID, embedded systems and wireless sensors, blockchain, AI, 3D printing, visualization tools and environmental technology. As DT applications are expected to increase in AEC, these technologies provide effective avenues for integration growth, particularly for PSC.

3.1.5.1. RFID technology. The RFID technology has attracted significant attention as a critical element for IoT. Hence, its consideration in the development of DT for PSC. RFID is an automatic identification technology that detects target objects through radio frequency signals for data processing [36]. Furthermore, it aids in tracking items and the retrieval of product-related information. The RFID technology uses a system of tags and readers to determine the status of targeted objects for data acquisition in real-time [52]. Due to the adaptability of RFID technology, it has been currently explored in tracking management systems and progress reporting in attempts to digitally twin the physical prefabrication components and processes [10,53]. In addition, RFID provides the opportunity to assign specific types of information to tags, and hence, data pertaining to emissions, management actions, quantity and productivity can be transmitted in the system.

3.1.5.2. Blockchain. The blockchain technology enables traceability, transparency and immutability throughout a fragmented or isolated chain of activities [54]. In this case, applying blockchain in DT for PSC builds trust in the data and information exchanged in the smart construction management network involving multiple stakeholders. The architecture of blockchain addresses the threat of having a single point of failure in the data exchange network system of DT for PSC and improves information visibility in collaborative environments for prefabrication [55]. By this, information transmitted in the DT system is more reliable and resilient. Blockchain provides a security framework in DT to enable smart contract management for suppliers/activities in modular construction supply chains. Therefore, information exchanged in DT for PSC has an added-value of accountability for activity-related information sharing in real-time [56].

3.1.5.3. Wireless sensors and embedded systems. Numerous wireless sensing technologies have been explored regarding DT for PSC based on functionalities and characteristics [57]. These sensors facilitate the co-creation of digital environments in real-time. This has enabled PSC products, objects and processes to be made *'smart'* via real-time data collection and decision-making [51]. More importantly, sensors have the crucial role of updating the digital environment with changes in the physical environment. Such timely information exchange is carried out in a wireless network. Currently, most of the sensing technologies for DT in PSC include GPS, laser sensors, pressure sensors and accelerations sensors. In application, the wireless sensors are attached or embedded into prefabricated components or processes to transmit data characterizing duration, form, location etc.

3.1.5.4. Artificial intelligence (AI). The increase in information from heterogenous sensors makes AI more effective in improving decision-making in DT at each stage of PSC. With AI, the DT for PSC can perform self-diagnosis, self-configuration and self-optimization, thereby influencing managerial interventions. These enriching capabilities of AI improve the accuracy and efficiency of technologies employed in smart prefabricated systems [58]. Additionally, connected streams of data from prefabrication processes and objects produce big data in real-time over a period in the DT system. Therefore, data-driven decision-making is required to generate intelligent assets and processes [59]. AI has been adopted in DT for PSC for purposes of predictive modelling, defect detection, simulation analysis and data segmentation [59,60].

3.1.5.5. 3D printing and robotics. The potential of 3D printing or additive manufacturing is fast making imprints in advancing smart industrialized construction or buildings. 3D printing technologies leverage on robotic systems and design/construction process-related data [61]. While the application of 3D digital fabrication is still an infant, it offers DT opportunities to improve the performance of robots employed in PSC in real-time [59]. For instance Ref. [62], demonstrated a robot's DT performing real-time monitoring and simulation at the construction activity level for motion planning and control. The virtual robots created in the DT system simultaneously planned the physical robot's trajectory and executed it in a real work environment [62]. Robotic systems aim to close the added-value deficits in productivity for prefabrication [61] and facilitate the autonomous creation of prefabricated modules or components. Therefore, having real-time data-driven robotic operations via DT enhances the synchronization and simulation of PSC processes.

3.1.5.6. Visualization technologies. Visualizing the information from sensors in prefabricated systems helps to better appreciate the state of real objects/components in real-time. This generates a comprehensive picture to complement the signals and outcomes from sensors and data in decision-making. With visualization technologies, DT can strengthen the management and control abilities for components/processes in PSC. Technologies such as virtual reality and IoT platforms perform visualized real-time monitoring of PSC activities (e.g. hoisting operations, circular economy, GHG) as decision support systems [9,63,64]. Data received in DT for PSC can be visualized as geometry or process state for decision analysis that involves simulation.

3.2. Critical review of DT in PSC

The critical review adopted the preferred reporting items for systematic reviews and meta-analysis (PRISMA) to examine empirical studies applying DT applications for (i) smart construction and (ii) carbon emissions in PSC. PRISMA has gained popularity because it incorporates two parts – systematic reviews and meta-analysis, for insights on a topic [65]. While the systematic review extracts evidence that fits a pre-defined eligibility criterion, the meta-analysis examines the outputs of pre-defined criteria from the individual studies for discussion. Steps in this process involve: (i) a target literature search; (ii) eligible case identification; (iii) data extraction [15].

3.2.1. Target literature search

A search was conducted within the initial Scopus database to extract empirical articles as the target literature. The abstracts and full texts of all articles in the initial search were carefully examined based on two criteria: (1) the article presented an empirical application of DT in PSC, and (2) the DT was used to evaluate carbon emissions in PSC. This process resulted in identifying 25 relevant articles for further examination.

3.2.2. Eligible case identification

The 25 selected articles reported on DT applications at the three main stages of PSC (i.e. production, transport and on-site assembly). Each of these PSC stages is referred to as a case application of DT in PSC for smart construction and carbon emissions. Therefore, 32 eligible cases were identified. Various variables pre-defined to be extracted from the eligible cases include emission sources, smart technologies and sensors used, information requirements, smart technology purpose and real-time transmission mode.

3.2.3. Data extraction

To extract these variables, a systematic review was conducted to record the details of the case applications respectively. Thereafter, meta-analysis was carried out on the identified variables from the case applications. The meta-analysis provides the quantitative outputs alongside the ways in which DT has been applied for carbon emissions at each stage of the PSC. It also allows findings to be compared among case applications.

4. Review results

The results of the critical review of DT case applications are presented in two parts – smart construction and carbon emissions.

4.1. DT in PSC for smart construction

Fig. 6 presents the results of various smart technologies used at each stage of the PSC when DT was explored. Generally, RFID has been the most explored smart technology in DT applications for PSC.

4.1.1. PSC production stage

With the production stage, smart technologies employed to facilitate DT include RFID, blockchain, laser scanning and robotics. As such, [57] used the RFID sensor system for real-time information status of individual prefabricated modules in modular construction to update virtual twin models. This ensured real-time progress monitoring of production activities. Besides progress monitoring, [52] employed the RFID system to capture location tracking and product identification data in panelized wood systems for material management. The instantaneous information gathered from RFID aids DT applications to be effective in planning resources amidst dynamic changes in prefabrication processes. In a different area, blockchain applications in DT have proven to provide better information traceability. As such, [55] linked a permissioned blockchain to an IoT platform to guarantee information origin for real-time changes in off-site production processes. Since DT typically has multiple sources of data, blockchain addresses the single point of failure problem in sensor networks and facilitates the integrity of actual modifications from multi-sources. For instance, the DT via blockchain can inspect transactions on prefabricated modules and invoke smart contracts based on trust. Quality assurance is crucial in prefabrication production. Hence, [66] used 3D laser scanning techniques for as-built data capturing to update the DT of prefabri-

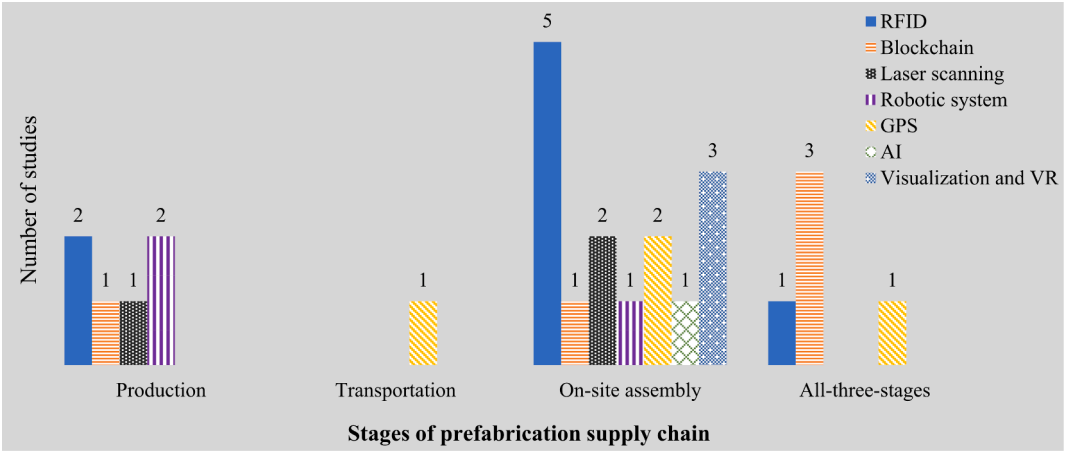


Fig. 6. Smart construction technologies for DT in PSC.

cated volumetric modules for geometric compliance. This improves structural integrity and detects or resolves assembly alignment problems. Robotics is essential to reduce or replace manual efforts in prefabrication production. [67] developed a collision-free workplace for active robots when producing modular units for construction. This was conducted by planning the paths of robots in DT. The robot's status information is transmitted to the virtual model, and a simulation of path planning is conducted for robotic operations. Likewise, via robotic fabrication and sensors, [68] gathered continuous information on building components for process monitoring in a DT platform. This led to an increase in automation and productivity concerning additive manufacturing.

4.1.2. PSC transportation stage

The transportation stage employed GPS technology in DT for smart construction. In this regard, [69] developed a DT for real-time logistics simulation to determine risks and route schedules. The GPS attached to a truck transmits information to the virtual replica in near real-time to ensure effective supply chain coordination.

4.1.3. PSC on-site assembly stage

The on-site assembly stage has witnessed the use of various smart technologies to promote DT in prefabrication. Due to uncertain outside conditions, [70] developed an intelligent modular construction system by converting on-site resources into smart objects via RFID and ultra-wide band (UWB) attachments. The RFID and UWB collected and integrated real-time data on location, identity and progress in the DT model. The high-fidelity DT model was used to monitor and transmit instructions to a robotic arm system for real-time interoperation of on-site resources and activities. This enhanced collaborative decision-making on robotic operations, automatic navigation and efficient resource management. In hoisting prefabricated components, [60] adopted RFID-long range (LoRa) technology to acquire data on the movement status of components in the hoisting process. By this, DT affords an innovative hoisting management system that reflects the current reality when monitoring the construction of prefabrication buildings. [71] combined RFID and GPS to make objects in construction *'smart'* for timely data acquisition on the component status and location. The sensory data transmitted to the DT makes progress monitoring effective for decision-making. Furthermore, [72] used RFID-LoRa for real-time status information of prefabricated modules on-site to monitor structural performance during installation in a digital replica. For on-site assembly services, [10] employed the RFID system to capture real-time data of on-site working processes in an IoT platform. Subsequently, the captured data is visualized with BIM-VR technologies to monitor progress, supervise daily operations and improve effective decisions for site management.

With intelligent technologies communicating ubiquitously in smart construction, integrity has to be guaranteed in the information that is shared. In this regard, [73] adopted a blockchain-based DT for real-time compliance checking to support accountable project-related information sharing. Particularly, using blockchain's time stamp function in DT. Normal data shared in DT is vulnerable to overwriting. Hence, engaging blockchain in DT improves traceability in data communication from sensor networks and adds confidence to data transactions from multiple stakeholders. From a different viewpoint [74], employed GPS to transmit the position information of modules to an IoT hoisting platform. This is to strengthen the management and intelligent control of multi-level hoisting processes in prefabrication.

In recent times, laser scanning technologies have been utilized to improve the quality of on-site prefabrication assembly. To accurately locate and assess building components during assembly, [75] carried out omnidirectional laser scanning of components to obtain point cloud (image) data for a DT platform. Then, the accurate positioning of components is monitored in the process for quality construction and safe site management. Similarly, [76] focused on the geometric quality assessment of prefabricated facades using laser scanning technology in DT. The laser scanner captured the as-built 3D representation of components in a point cloud and reconstructed digital replicas in the DT platform. Such 3D semantic models in DT could be used to detect geometric errors and improve quality assessment. Another DT application in smart prefabricated construction focused on using AI for safety. As such [63], applied AI in DT to predict safety risks for hoisting prefabricated components based on real-time site data gathered in the virtual model. In addition, [77] used visualization techniques in DT for real-time interactions of construction objects to improve decision analysis con-

cerning on-site safety risk management when conducting hoisting activities. [9] developed a VR system that identified prefabricated components used in real-time for construction buildings on sites. The added value of this digital replica is identifying and visualizing the circularity potential of materials used in the building.

4.1.4. All-three-stages of PSC

Some studies have made attempts to use certain smart technologies throughout the three stages of PSC in developing a DT. For instance, Ref. [53] adopted the RFID technology for real-time information and visibility of prefabricated modules from the off-site factory through transportation to site assembly. Progress monitoring was achievable in this study by attaching RFID sensors to modules, and process data was transmitted to the DT at each stage of the supply chain. By this, collaborative decision-making is enhanced for inventory management at production, location status for transportation and on-site asset and quality management. Furthermore, the potential of blockchain to boost confidence in real-time information shared among PSC stakeholders has been realized in DT. [54] introduced a permissioned blockchain in an IoT-BIM platform to offer secured interoperability of information/data transmitted from RFID technologies in the supply chain of modular units. The security in information transmission increases trust and privacy for various PSC stakeholders. Similarly, [56] used blockchain for smart modular construction by tackling the *"islands of information"* problem which arises when different stakeholders across PSC interact in real-time. Using a DT platform, the blockchain facilitated data on real-time progress of orders and materials from multiple stakeholders such as clients, module manufacturers and logistics company. Additionally, immutability in construction progress traceability and monitoring is enhanced for smart construction. Another blockchain-based DT application for smart construction is smart contracts. As such, [78] developed a blockchain system component status sharing for PSC stakeholders. This is to ensure the integrity of information transmitted in DT for smart contract applications. To trace the location of modular units throughout PSC, [57] engaged GPS for real-time automatic data collection on the prefabrication delivery process. Within the DT platform, the GPS facilitated traceability and adjustments for effective decisions to be made. Fig. 7 provides a summary of the findings on smart technologies for DT in PSC alongside the information requirements.

4.2. DT in PSC for carbon emissions

The findings from the critical review showed several smart technologies for estimating carbon emissions in PSC. Fig. 8 provides the results of smart technologies employed to evaluate real-time carbon emissions in DT at each stage of the PSC.

4.2.1. PSC production stage

Various materials and processes are usually employed in the production stage. Therefore, smart technologies have to be carefully considered to ensure real-time capturing of carbon emissions in DT platforms. To capture carbon emissions in DT, the most critical action is to identify and replicate the emission sources in the DT. In the study by Ref. [36]; the RFID system and proximity switch sensors were employed for emission sources involving raw materials and workstation equipment respectively. The RFID system tagged basic information and embodied carbon emissions to components (i.e. steel and concrete) for data transmission in the DT. For machines used in the production line, the proximity switch sensors gathered time/duration related information to calculate emissions at various workstations in real-time. These calculations were based on emissions coefficients derived from established emissions databases. Furthermore, [35] used the RFID system to connect the composition of materials in a component for component information identification. By this, embodied emissions can be assessed in the DT. With a different technique, [35] adopted laser photoelectric sensors for real-time component location and processing time when a component enters and leaves the processing station. The time difference serves as the basis for emissions in the DT computing platform based on the energy consumption data at the workstation.

4.2.2. PSC transportation stage

GPS has been the main technology for tracking distance in construction prefabrication haulage. [36] applied GPS alongside smart applications on vehicles transporting prefabrication components. Tracking data was collected from the GPS and energy consumed by the vehicle per distance travelled was computed in the DT platform in real-time. Consequently, the carbons emitted can be generated for the transportation process instantaneously.

4.2.3. PSC on-site stage

The sources of emissions for on-site activities are mostly a result of machines and movements. Three types of construction equipment have been explored for real-time carbon emissions on-site for DT. The first relates to the tower crane. In this regard, [36] introduced acceleration sensors to estimate the working time of cranes by measuring the motion state of the tower rope. The data gathered from the sensor is transferred to a DT platform in real-time. Likewise, acceleration sensors were also employed by Ref. [64] to capture the motion state of the crane's boom and hook to determine the running time for the actual operation duration. The second piece of equipment is the construction elevator. Pressure sensors were adopted to monitor the movement of elevators for data gathering and real-time transmission [36]. By using the pressure difference method, the pressure sensors measure height through the change of pressure [64]. The resulting data from these pressure sensors are shared in a network to a DT platform for the corresponding carbon emissions to be computed. The third equipment monitored for carbon emissions was on-site transfer vehicles. As such, [36,64] attached GPS to transfer vehicles moving prefabrication components on-site. The GPS detected the position of transfer vehicles (material handling equipment) and transmitted the working time of the vehicle to the DT platform for carbon emissions assessments. Fig. 9 summarizes smart technologies employed for real-time information transmission from materials, equipment and processes at each stage of the PSC regarding DT application.

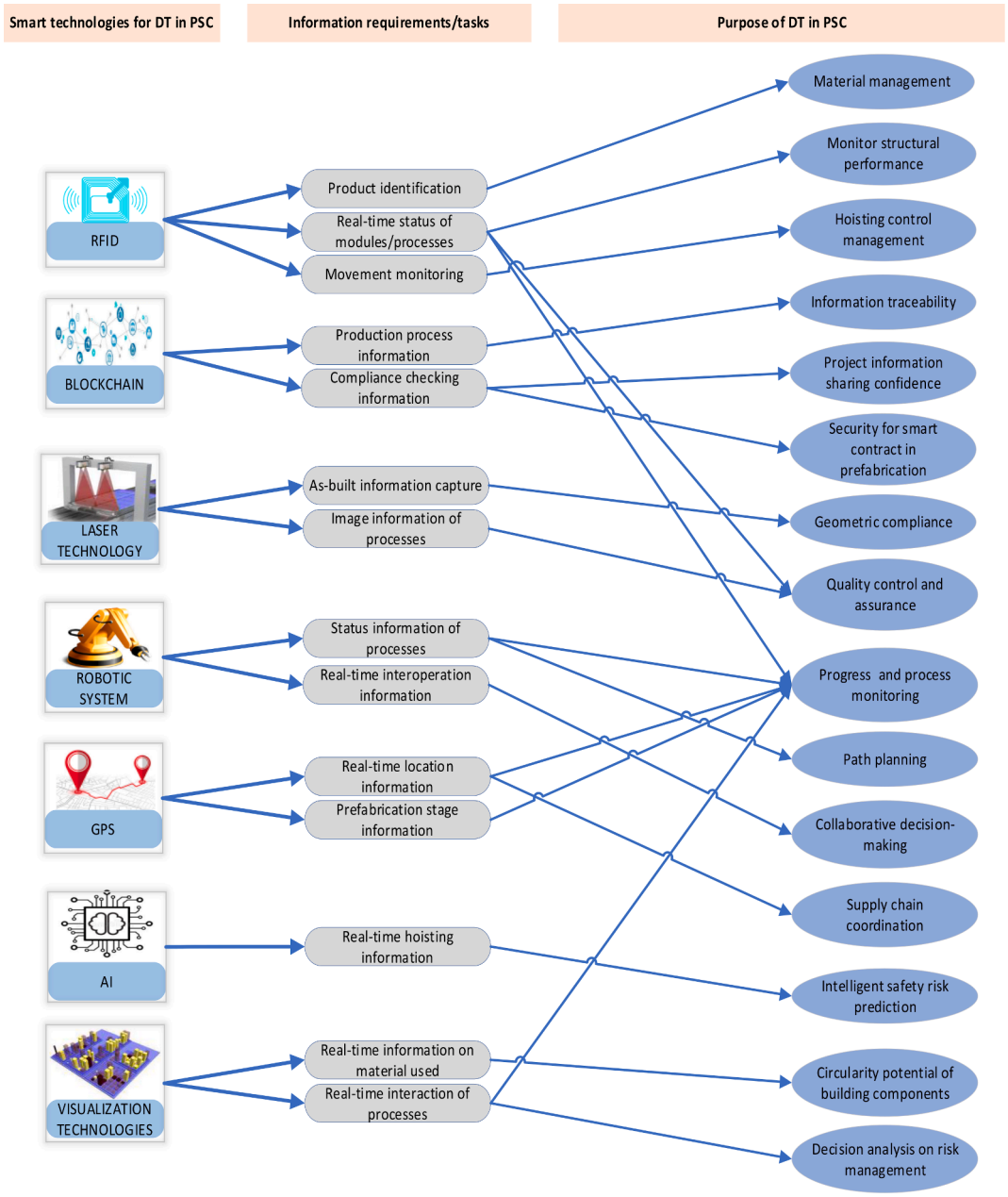


Fig. 7. Smart technologies and their purposes in DT for PSC.

5. Discussions and future research directions

This study highlighted the current issues requiring critical attention to develop DT for evaluating the carbon emissions of PSC and the corresponding research directions. The contribution of the paper lies in applying both quantitative and qualitative analyses on the literature of DT-based smart technologies that investigated the potential of carbon emissions evaluation in PSC. In this regard, five critical issues are discussed as follows for the advancement of DT in PSC.

5.1. Increase in smart construction objects

In order to promote DT applications to be widely adopted in prefabricated construction practices, increased awareness and attention are needed to make the PSC process smarter. Thus, transforming products and processes in PSC into smart construction objects (SCOs). With SCOs, products and processes are equipped with awareness and information transfer capabilities. Currently, efforts to transform prefabrication products and processes into smart objects are at the small task level, involving few smart technologies. However, research on how these smart-enabling technologies would be integrated across different PSC stages for resources (e.g. machin-

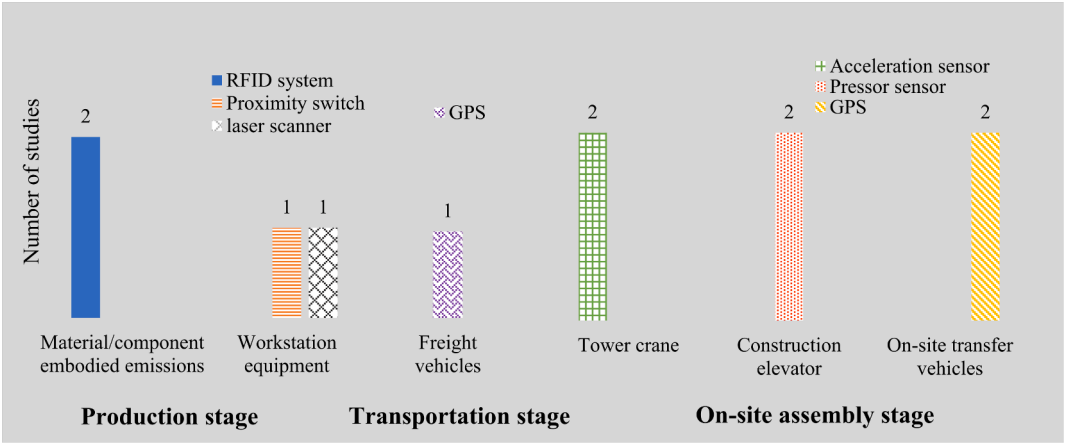


Fig. 8. Application of DT in PSC for carbon emissions.

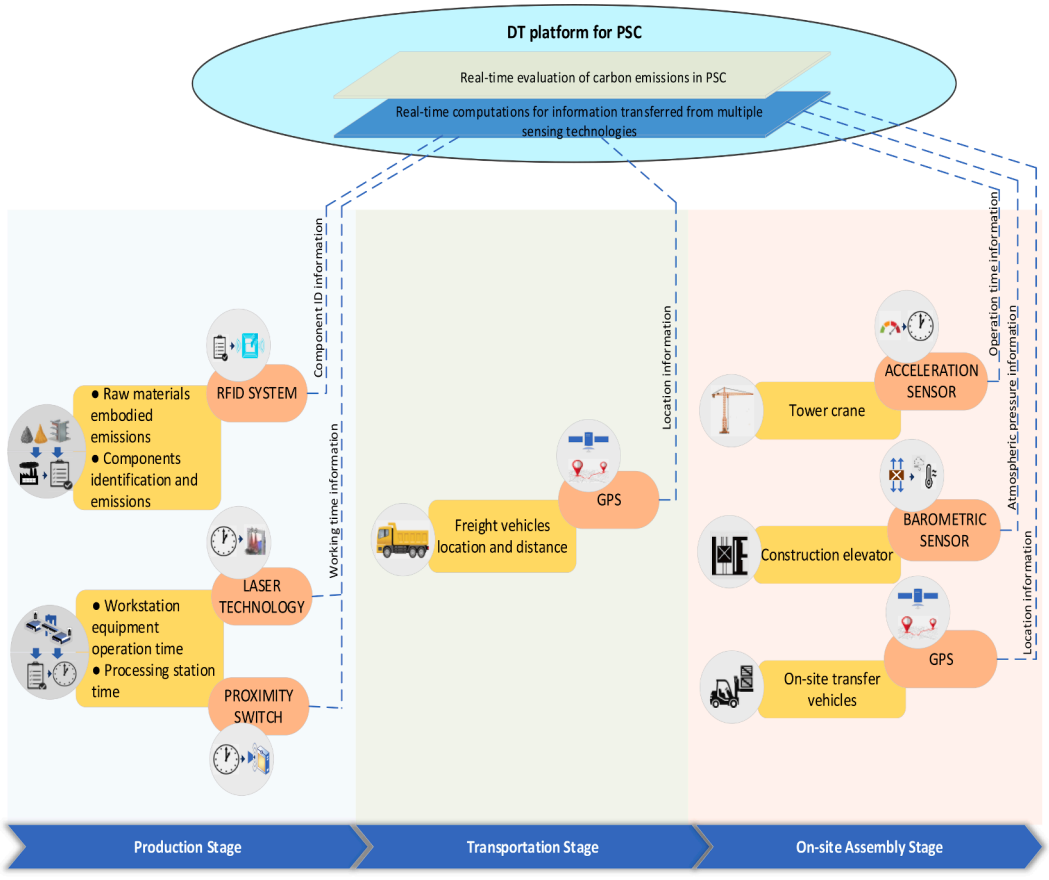


Fig. 9. Smart technologies used in DT for carbon emissions at the three stages of prefabrication.

ery and materials) remains nascent and uncertain. As revealed in the science mapping analysis, automatic monitoring systems were identified as one of the main clusters in DT for PSC literature. This supports the potential of SCOs to facilitate data transfer and monitoring in DT. SCOs confer autonomy, awareness and interactive data abilities to the prefabrication processes and products they represent [51]. Hence, SCOs facilitate autonomous exchange of real-time information in DT as products and processes interact. Albeit harnessing SCOs with processes and products within PSC presents problems that require further investigation to ensure their effectiveness for DT. First, coordinating information transfer from multiple SCOs as they interact within and across PSC in real-time raises concerns about data authenticity. Data authenticity has been identified as a crucial element in information exchange, particularly DT [79]. This is because distorted information from the physical asset can mislead decision-making in construction management. There-

fore, integrating blockchain applications into SCOs information flow may help manufacturers, suppliers, transporters and project managers to build trust in sharing interrelated data of their processes and products in the DT platform. While this trajectory of using blockchain holds opportunities for improving trust in real-time data transfer, it may cause data transfer delays, particularly when the PSC is large and involves multi-complex sensor devices for the building. In this regard, further research should focus on developing blockchain authentication networks for activities of each PSC stage to mitigate delays in SCOs data transfer in DT. Second, exploratory research would still be needed to intensify how physical processes or products in PSC should be effectively represented in DT. With this, future research should provide frameworks or taxonomies detailing smart technologies that efficiently represent common prefabrication activities and products at production, transportation and on-site assembly stages. The smart technology's ability to represent geometry (shape) and/or motion (status) on their respective prefabrication processes and products must be highlighted to enrich knowledge on DT for PSC. Lastly, further research is required to make SCOs more robust for opportunities in intelligent construction and buildings.

5.2. Accuracy of DT in replicating emissions from physical activities/objects

Research on DT for carbon emissions in PSC is emerging in construction literature. Few attempts in this research agenda have raised both difficulties and opportunities when considering DT as a tool for evaluating emissions. One of the main difficulties relates to the accuracy of smart technologies in detecting calibrated processes and products. For instance, the RFID strength influences its precision and is affected by metal surfaces in the working environment. Also, vibrations in the production environment can cause laser sensor deviations, causing problems in their ability to capture work processes. This impacts the reliability of emissions information transferred to the DT. Other sensor applications (e.g. phone GPS) used to monitor emissions at the transportation stage of prefabrication had human control limitations, making them prone to errors. In this regard, future research should focus on developing appropriate methods and tolerance levels to improve information reliability for smart technologies used for DT in PSC. Another issue is the weak link between information parameters derived from sensing technologies on physical assets/objects and the carbon emissions databases or inventories. Different equipment, processes and materials may require different parameters such as operating time and distance for carbon emissions computations. Furthermore, the local production setting may pose challenges to the accurate calibration of processes regarding emissions coefficients. Therefore, future research should focus on developing strong theoretical foundations for linking emissions coefficients to various parameters of processes and equipment on a continuous basis in PSC. This theoretical lens can be used for creating system architecture for an efficient online system covering a variety of parameters.

5.3. Integrating AI into real-time camera capture for DT

The use of cameras in PSC working environments is increasing for purposes of security, worker safety and productivity monitoring. However, its potential as a sensing technology for data transfer in DT has been under-researched in the literature. The ability of cameras to capture object motions in real-time and provide visual identification of objects would be helpful for AI to compute, calibrate and programme activities in PSC for DT. For instance, AI can be incorporated into the real-time video images derived from various cameras in the production line to determine two attributes: (i) the motion state of products/materials, and (ii) the type of product/material. By this, developing an AI-based platform to compute and categorize methods for capturing the journey of products in the production line with live video images would be a fertile avenue for future research in DT. This approach offers new opportunities for acquiring data on the real state of practice at the production stage of PSC. Such data from the AI-camera integration (i.e. motion state and type of product), can be subsequently converted for carbon emissions evaluations when applying DT in PSC. Furthermore, with AI-camera integration, geometric quality can be assessed in real-time at the production and on-site assembly stages. The real-time video images of as-produced prefabricated products can be compared with the real-time on-site video images of products and the AI-based computing platform would determine quality compliance. In addition, the integration of AI into real-time video cameras has great potential for robotic applications at the production and on-site assembly stages.

5.4. Standardizing the data structure for DT in PSC

The type of data to be transferred in DT for PSC still needs more research to strengthen data acquisition. Recent studies concerning DT for PSC have highlighted diverse ways of acquiring data from some processes, machinery and materials used in PSC for DT. While this development underscores alternatives to data acquisition, it raises concerns about the type and structure of data that would be effective for DT applications. Different machinery and processes would require different measurement units to acquire real-time data. Therefore, future research can focus on providing standardized methods of structuring real-time data derived from different processes, machines and materials for DT applications in PSC. This would enable a standardized structure of collecting and transferring information for DT computations in PSC such as carbon emissions. Also, such efforts would spur research to develop customized DT platforms for PSC stages, hence promoting the use of DT in PSC.

5.5. Two-way data transmission in DT

Most of the studies applying DT in PSC concentrated on one-way data transmission. Thus, information transfers from sensing technologies (physical assets) to DT platforms. However, one main strength of DT is transferring instructions or decisions from the DT platform to the sensing technologies (physical assets) [62]. This results in two-way data transmission. With this gap in literature, future studies should focus on developing a mechanism for two-way data flow as the key feature of the DT model to the physical asset at any stage of PSC in real-time (e.g. Ref. [70]. This would help instructions to be transferred from DT to physical assets to enable real-time interventions when monitoring carbon emissions. Fig. 10 provides the directions for future research.

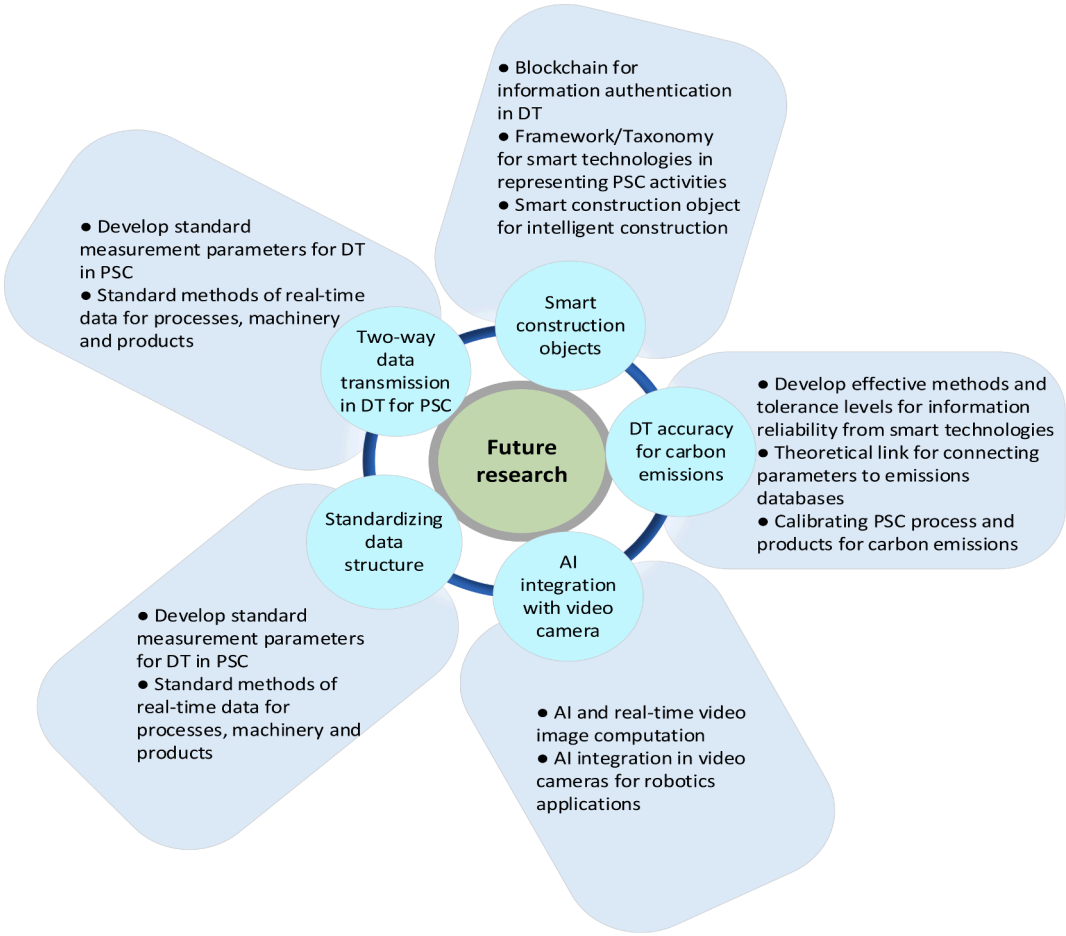


Fig. 10. Future research directions for DT in PSC.

6. Theoretical and practical implications

The findings in this study have implications for research and practice. For research, the insights presented in the five research directions for DT in PSC provide a firm foundation for the formulation and development of theoretical frameworks/models for real-time carbon emissions monitoring. Theoretically, this study expands the discussions on emissions in construction literature to a new niche (i.e. prefabrication) for the possibility of real-time monitoring. Such theoretical frameworks spurred by this study would enable researchers to develop novel approaches suitable for capturing processes in PSC, and hence, advance the understanding and application of DT in prefabrication.

Instead of one-off emissions calculations usually conducted by practitioners at the design stage of buildings, this study offers opportunities for continuous evaluation of carbon emissions in real-time for prefabrication. For practitioners, this would be useful in making timely decisions at each stage of the prefabrication process for appropriate interventions to be carried out. In this regard, project managers and policy makers would be able to assess the real emissions associated with prefabrication when carbon reduction targets or offsets are part of the project goals. Furthermore, the DT in PSC approached revealed in this study would provide practitioners with relevant knowledge on the information requirements needed from smart technologies for real-time evaluations of emissions.

7. Conclusions

This study presents the findings of a comprehensive review of DT applications in PSC to advance smart construction and carbon emission evaluation using the mixed-methods approach. From the scientometric analysis, the key areas of research interests regarding DT for PSC relate to emissions and energy control, AI-based decision-making, automatic monitoring, visualization and blockchain integration. For research collaboration, China and the UK have collaborated the most with other countries to explore DT in PSC. In the critical review, smart construction technologies predominantly applied in DT at each stage of the PSC for data real-time data acquisition were RFID and GPS. 3D laser scanning technologies are gaining traction for physical asset data capture and these technologies could be beneficial in the initiation of DT in PSC. Furthermore, the critical review revealed that smart technologies such as RFID proximity switch, laser sensors, pressure sensors and acceleration sensors are more suitable for representing real-time carbon emissions of materials and processes at each stage of prefabrication in a DT platform. These smart technologies have exposed new ways to collect

real-time information status and operating time of various machines, processes and materials at the production and on-site assembly for emissions computations in DT. For the transportation stage, GPS was the main sensing technology for real-time information collection in DT for emissions computations. The smart technologies identified in the study offer a comprehensive approach to develop and implement DT in PSC. Particularly, the information requirements needed from these technologies to evaluate carbon emissions at each stage of the PSC would motivate researchers and practitioners on how to improve real-time monitoring of processes and materials in prefabricated construction.

With an emerging interest to advance DT at different stages of the PSC, several gaps and weaknesses were identified in extant literature. In that regard, five fertile avenues to advance knowledge and accelerate DT applications in PSC were provided in this study. Thus, increasing SCOs in prefabrication, improving accurate replication of emissions in DT, integrating AI into real-time video cameras, standardizing data structure for DT and exploring two-way data transmission in DT for PSC. This study makes significant contributions to the body of knowledge on DT by: (i) revealing five core areas of interest and the crucial areas that are under-represented as DT for PSC literature is emerging, (ii) providing deep insights on how smart technologies can be applied for carbon emissions monitoring in DT, and (iii) offering effective avenues for future research to improve and consolidate DT applications in PSC. The outcomes of the study would aid researchers in developing principles for standardizing information requirements and measurement units for prefabrication processes, machines and materials in DT for various purposes. Nonetheless, this study has the limitation of considering English articles only. Also, another limitation is DT and its associated technologies are rapidly evolving. Hence, smart technologies that are about to enter the construction market may be captured in the review. Therefore, researchers and industry practitioners should closely monitor these developments and integrate smart technologies that add value over time. However, the insights presented in this study hold great potential for future research to advance the knowledge development on DT for PSC.

Declaration of competing interest

The authors declare that they have no known competing interests.

Data availability

Data will be made available on request.

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References

[1] A. Prabhakaran, A.M. Mahamadu, L. Mahdjoubi, Understanding the challenges of immersive technology use in the architecture and construction industry: a systematic review, Autom. ConStruct. 137 (2022) 104228, https://doi.org/10.1016/j.autcon.2022.104228.
[2] A.J. Sánchez-Garrido, I.J. Navarro, J. García, V. Yepes, A systematic literature review on modern methods of construction in building: an integrated approach using machine learning, J. Build. Eng. (2023) 106725.
[3] Z. Zheng, F. Wang, G. Gong, H. Yang, D. Han, Intelligent technologies for construction machinery using data-driven methods, Autom. ConStruct. 147 (2023) 104711, https://doi.org/10.1016/j.autcon.2022.104711.
[4] D.G. Broo, M. Bravo-Haro, J. Schooling, Design and implementation of a smart infrastructure digital twin, Autom. ConStruct. 136 (2022) 104171, https://doi.org/10.1016/j.autcon.2022.104171.
[5] G.B. Ozturk, Digital twin research in the AECO-FM industry, J. Build. Eng. 40 (2021) 102730, https://doi.org/10.1016/j.jobe.2021.102730.
[6] J. Zhao, H. Feng, Q. Chen, B.G. de Soto, Developing a conceptual framework for the application of digital twin technologies to revamp building operation and maintenance processes, J. Build. Eng. 49 (2022) 104028, https://doi.org/10.1016/j.jobe.2022.104028.
[7] I. Osadcha, A. Jurelionis, P. Fokaides, Geometric parameter updating in digital twin of built assets: a systematic literature review, J. Build. Eng. (2023) 106704.
[8] X. Wang, Q. Du, C. Lu, J. Li, Exploration in carbon emission reduction effect of low-carbon practices in prefabricated building supply chain, J. Clean. Prod. 368 (2022) 133153, https://doi.org/10.1016/j.jclepro.2022.133153.
[9] T.M. O'Grady, N. Brajkovich, R. Minunno, H.Y. Chong, G.M. Morrison, Circular economy and virtual reality in advanced BIM-based prefabricated construction, Energies 14 (13) (2021) 4065, https://doi.org/10.3390/en14134065.
[10] C.Z. Li, F. Xue, X. Li, J. Hong, G.Q. Shen, An Internet of Things-enabled BIM platform for on-site assembly services in prefabricated construction, Autom. ConStruct. 89 (2018) 146–161, https://doi.org/10.1016/j.autcon.2018.01.001.
[11] J.S. Goulding, F. Pour Rahimian, M. Arif, M.D. Sharp, New offsite production and business models in construction: priorities for the future research agenda, Architect. Eng. Des. Manag. 11 (3) (2015) 163–184, https://doi.org/10.1080/17452007.2014.891501.
[12] V. Tavares, N. Soares, N. Raposo, P. Marques, F. Freire, Prefabricated versus conventional construction: comparing life-cycle impacts of alternative structural materials, J. Build. Eng. 41 (2021) 102705.
[13] Y. Tian, S. Spatari, Environmental life cycle evaluation of prefabricated residential construction in China, J. Build. Eng. 57 (2022) 104776.
[14] J. Jayawardana, M. Sandanayake, J.A.S.C. Jayasinghe, A.K. Kulatunga, G. Zhang, A comparative life cycle assessment of prefabricated and traditional construction–A case of a developing country, J. Build. Eng. 72 (2023) 106550.
[15] Y. Teng, K. Li, W. Pan, T. Ng, Reducing building life cycle carbon emissions through prefabrication: evidence from and gaps in empirical studies, Build. Environ. 132 (2018) 125–136, https://doi.org/10.1016/j.buildenv.2018.01.026.
[16] S.K. Yevu, E.K. Owusu, A.P. Chan, K. Oti-Sarpong, I.Y. Wuni, M.O. Tetteh, Systematic Review on the Integration of Building Information Modelling and Prefabrication Construction for Low-Carbon Building Delivery, Building Research & Information, 2022, pp. 1–22, https://doi.org/10.1080/09613218.2022.2131504.
[17] F. Jiang, L. Ma, T. Broyd, K. Chen, Digital twin and its implementations in the civil engineering sector, Autom. ConStruct. 130 (2021) 103838, https://doi.org/10.1016/j.autcon.2021.103838.
[18] S.M. Sepasgozar, Digital twin and cities, in: The Palgrave Encyclopedia of Urban and Regional Futures, Springer International Publishing, Cham, 2022, pp. 1–6, https://doi.org/10.1007/978-3-030-51812-7_253-1.
[19] M. Grieves, Digital Twin: manufacturing excellence through virtual factory replication, White Paper. https://www.3ds.com/fileadmin/PRODUCTS-SERVICES/DELMIA/PDF/Whitepaper/DELMIA-APRISO-Digital-Twin-Whitepaper.pdf, 2014.
[20] M. Grieves, Origins of the Digital Twin Concept, 2016, https://doi.org/10.13140/RG.2.2.26367.61609.

[21] J. Lee, E. Lapira, B. Bagheri, H.A. Kao, Recent advances and trends in predictive manufacturing systems in big data environment, Manuf. lett. 1 (1) (2013) 38–41, https://doi.org/10.1016/j.mfglet.2013.09.005.
[22] J. Ríos, J.C. Hernandez, M. Oliva, F. Mas, Product avatar as digital counterpart of a physical individual product: literature review and implications in an aircraft, Transdiscipl. Lifecycle Anal. Syst. (2015) 657–666, https://doi.org/10.3233/978-1-61499-544-9-657.
[23] A. Canedo, Industrial IoT Lifecycle via Digital Twins, ACM Press, 2016. https://dl.acm.org/doi/10.1145/2968456.2974007.
[24] M. Grieves, Virtually Intelligent Product Systems: Digital and Physical Twins. Complex Systems Engineering: Theory and Practice, 2019, pp. 175–200, https://doi.org/10.2514/5.9781624105654.0175.0200.
[25] F. Tao, M. Zhang, A.Y.C. Nee, Digital Twin Driven Smart Manufacturing, Academic Press, London. UK, 2019, https://doi.org/10.1016/C2018-0-02206-9.
[26] R. Sacks, I. Brilakis, E. Pikas, H.S. Xie, M. Girolami, Construction with digital twin information systems, Data Cent. Eng. 1 (2020) 1–26, https://doi.org/10.1017/dce.2020.16.
[27] RICS, Digital Twin from Design to Handover of Constructed Assets, 2022. https://www.rics.org/globalassets/wbef-website/reports-and-research/digital-twins-from-design-to-handover-of-constructed-assets.pdf. Retrieved 06/09/2022.
[28] S.A. Talmaki, V.R. Kamat, Sensor acquisition and allocation for real-time monitoring of articulated construction equipment in digital twins, Sensors 22 (19) (2022) 7635, https://doi.org/10.3390/s22197635.
[29] P. Teicholz (Ed.), BIM for Facility Managers, John Wiley & Sons, New Jersey, 2013 978-1-118-38281-3.
[30] S.M. Sepasgozar, Differentiating digital twin from digital shadow: elucidating a paradigm shift to expedite a smart, sustainable built environment, Buildings 11 (4) (2021) 151, https://doi.org/10.3390/buildings11040151.
[31] A. Bolton, L. Butler, I. Dabson, M. Enzer, M. Evans, T. Fenemore, F. Harradence, E. Keaney, A. Kemp, A. Luck, N. Pawsey, S. Saville, J. Schooling, M. Sharp, T. Smith, J. Tennison, J. Whyte, A. Wilson, C. Makri, The Gemini Principles, Apollo University of Cambridge Repository, Cambridge, UK, 2018, https://doi.org/10.17863/CAM.32260.
[32] C. Goodier, A. Gibb, Future opportunities for offsite in the UK, Construct. Manag. Econ. 25 (6) (2007) 585–595, https://doi.org/10.1080/01446190601071821.
[33] M. Wasim, T.M. Han, H. Huang, M. Madiyev, T.D. Ngo, An approach for sustainable, cost-effective and optimised material design for the prefabricated non-structural components of residential buildings, J. Build. Eng. 32 (2020) 101474.
[34] B. Qi, M. Razkenari, A. Costin, C. Kibert, M. Fu, A systematic review of emerging technologies in industrialized construction, J. Build. Eng. 39 (2021) 102265.
[35] X. Tao, C. Mao, F. Xie, G. Liu, P. Xu, Greenhouse gas emission monitoring system for manufacturing prefabricated components, Autom. ConStruct. 93 (2018) 361–374, https://doi.org/10.1016/j.autcon.2018.05.015.
[36] G. Liu, R. Chen, P. Xu, Y. Fu, C. Mao, J. Hong, Real-time carbon emission monitoring in prefabricated construction, Autom. ConStruct. 110 (2020) 102945, https://doi.org/10.1016/j.autcon.2019.102945.
[37] M. Robati, P. Oldfield, A.A. Nezhad, D.G. Carmichael, A. Kuru, Carbon value engineering: a framework for integrating embodied carbon and cost reduction strategies in building design, Build. Environ. 192 (2021) 107620, https://doi.org/10.1016/j.buildenv.2021.107620.
[38] M. Sandanayake, G. Zhang, S. Setunge, Estimation of environmental emissions and impacts of building construction–A decision making tool for contractors, J. Build. Eng. 21 (2019) 173–185.
[39] S.O. Ajayi, L.O. Oyedele, O.M. Ilori, Changing significance of embodied energy: a comparative study of material specifications and building energy sources, J. Build. Eng. 23 (2019) 324–333.
[40] R. Akram, M.J. Thaheem, A.R. Nasir, T.H. Ali, S. Khan, Exploring the role of building information modeling in construction safety through science mapping, Saf. Sci. 120 (2019) 456–470, https://doi.org/10.1016/j.ssci.2019.07.036.
[41] W.S. Alaloul, A.H. Qureshi, M.A. Musarat, S. Saad, Evolution of close-range detection and data acquisition technologies towards automation in construction progress monitoring, J. Build. Eng. 43 (2021) 102877.
[42] S.K. Yevu, T.W. Ann, A. Darko, Digitalization of construction supply chain and procurement in the built environment: emerging technologies and opportunities for sustainable processes, J. Clean. Prod. 322 (2021) 129093, https://doi.org/10.1016/j.jclepro.2021.129093.
[43] C. Chen, Science mapping: a systematic review of the literature, J. data inform. sci. 2 (2) (2017) 1–40, https://doi.org/10.1515/jdis-2017-0006.
[44] M. Al Hattab, The dynamic evolution of synergies between BIM and sustainability: a text mining and network theory approach, J. Build. Eng. 37 (2021) 102159.
[45] D.G.J. Opoku, S. Perera, R. Osei-Kyei, M. Rashidi, Digital twin application in the construction industry: a literature review, J. Build. Eng. 40 (2021) 102726, https://doi.org/10.1016/j.jobe.2021.102726.
[46] L.A. Martins, V. Soebarto, T. Williamson, A systematic review of personal thermal comfort models, Build. Environ. 207 (2022) 108502, https://doi.org/10.1016/j.buildenv.2021.108502.
[47] N.J. van Eck, L. Waltman, Visualizing bibliometric networks, in: Y. Ding, R. Rousseau, D. Wolfram (Eds.), Measuring Scholarly Impact: Methods and Practice, Springer International Publishing, Cham, 2014, pp. 285–320, https://doi.org/10.1007/978-3-319-10377-8_13.
[48] L. Liao, L. Quan, C. Yang, L. Li, Knowledge synthesis of intelligent decision techniques applications in the AECO industry, Autom. ConStruct. 140 (2022) 104304, https://doi.org/10.1016/j.autcon.2022.104304.
[49] N.J. van Eck, L. Waltman, Software survey: VOSviewer, a computer program for bibliometric mapping, Scientometrics 84 (2) (2010) 523–538, https://doi.org/10.1007/s11192-009-0146-3.
[50] T. Pullen, D.M. Hall, J. Lessing, A Preliminary Overview of Emerging Trends for Industrialized Construction in the United States, 2019, https://doi.org/10.3929/ethz-b-000331901.
[51] Y. Niu, W. Lu, K. Chen, G.G. Huang, C. Anumba, Smart construction objects, J. Comput. Civ. Eng. 30 (4) (2016) 04015070, https://doi.org/10.1061/(ASCE)CP.1943-5487.0000550.
[52] M. Wang, M.S. Altaf, M. Al-Hussein, Y. Ma, Framework for an IoT-based shop floor material management system for panelized homebuilding, Int. J. Construct. Manag. 20 (2) (2020) 130–145, https://doi.org/10.1080/15623599.2018.1484554.
[53] R.Y. Zhong, Y. Peng, F. Xue, J. Fang, W. Zou, H. Luo, G.Q. Huang, Prefabricated construction enabled by the Internet-of-Things, Autom. ConStruct. 76 (2017) 59–70, https://doi.org/10.1016/j.autcon.2017.01.006.
[54] X. Li, W. Lu, F. Xue, L. Wu, R. Zhao, J. Lou, J. Xu, Blockchain-enabled IoT-BIM platform for supply chain management in modular construction, J. Construct. Eng. Manag. 148 (2) (2022) 04021195, https://doi.org/10.1061/(ASCE)CO.1943-7862.0002229.
[55] L. Wu, W. Lu, F. Xue, X. Li, R. Zhao, M. Tang, Linking permissioned blockchain to Internet of Things (IoT)-BIM platform for off-site production management in modular construction, Comput. Ind. 135 (2022) 103573, https://doi.org/10.1016/j.compind.2021.103573.
[56] Y. Jiang, X. Liu, K. Kang, Z. Wang, R.Y. Zhong, G.Q. Huang, Blockchain-enabled cyber-physical smart modular integrated construction, Comput. Ind. 133 (2021) 103553, https://doi.org/10.1016/j.compind.2021.103553.
[57] Y. Zhai, K. Chen, J.X. Zhou, J. Cao, Z. Lyu, X. Jin, G.Q. Huang, An Internet of Things-enabled BIM platform for modular integrated construction: a case study in Hong Kong, Adv. Eng. Inf. 42 (2019) 100997, https://doi.org/10.1016/j.aei.2019.100997.
[58] M. Wang, C.C. Wang, S. Sepasgozar, S. Zlatanova, A systematic review of digital technology adoption in off-site construction: current status and future direction towards industry 4.0, Buildings 10 (11) (2020) 204, https://doi.org/10.3390/buildings10110204.
[59] R. He, M. Li, V.J. Gan, J. Ma, BIM-enabled computerized design and digital fabrication of industrialized buildings: a case study, J. Clean. Prod. 278 (2021) 123505, https://doi.org/10.1016/j.jclepro.2020.123505.
[60] Y. Zhao, C. Cao, Z. Liu, A framework for prefabricated component hoisting management systems based on digital twin technology, Buildings 12 (3) (2022) 276, https://doi.org/10.3390/buildings12030276.
[61] B. Xiao, C. Chen, X. Yin, Recent advancements of robotics in construction, Autom. ConStruct. 144 (2022) 104591, https://doi.org/10.1016/j.autcon.2022.104591.
[62] C.J. Liang, W. McGee, C.C. Menassa, V.R. Kamat, Real-time state synchronization between physical construction robots and process-level digital twins, Construct. Robot. 6 (1) (2022) 57–73, https://doi.org/10.1007/s41693-022-00068-1.

[63] Z.S. Liu, X.T. Meng, Z.Z. Xing, C.F. Cao, Y.Y. Jiao, A.X. Li, Digital twin-based intelligent safety risks prediction of prefabricated construction hoisting, Sustainability 14 (9) (2022) 5179, https://doi.org/10.3390/su14095179.
[64] G. Liu, H. Yang, Y. Fu, C. Mao, P. Xu, J. Hong, R. Li, Cyber-physical system-based real-time monitoring and visualization of greenhouse gas emissions of prefabricated construction, J. Clean. Prod. 246 (2020) 119059, https://doi.org/10.1016/j.jclepro.2019.119059.
[65] D. Moher, A. Liberati, J. Tetzlaff, D.G. Altman, Prisma Group., Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement, Int. J. Surg. 8 (5) (2010) 336–341, https://doi.org/10.1016/j.ijsu.2010.02.007.
[66] C. Rausch, R. Lu, S. Talebi, C. Haas, Deploying 3D scanning based geometric digital twins during fabrication and assembly in offsite manufacturing, Int. J. Construct. Manag. (2021) 1–14, https://doi.org/10.1080/15623599.2021.1896942.
[67] Q. Zhu, T. Zhou, P. Xia, J. Du, Robot planning for active collision avoidance in modular construction: pipe skids example, J. Construct. Eng. Manag. 148 (10) (2022) 04022114, https://doi.org/10.1061/(ASCE)CO.1943-7862.0002374.
[68] S. Bodea, P. Mindermann, G.T. Gresser, A. Menges, Additive manufacturing of large coreless filament wound composite elements for building construction, 3D Print. Addit. Manuf. 9 (3) (2022) 145–160, https://doi.org/10.1089/3dp.2020.0346.
[69] D. Lee, S. Lee, Digital twin for supply chain coordination in modular construction, Appl. Sci. 11 (13) (2021) 5909, https://doi.org/10.3390/app11135909.
[70] Y. Jiang, M. Li, D. Guo, W. Wu, R.Y. Zhong, G.Q. Huang, Digital twin-enabled smart modular integrated construction system for on-site assembly, Comput. Ind. 136 (2022) 103594, https://doi.org/10.1016/j.compind.2021.103594.
[71] J.X. Zhou, G.Q. Shen, S.H. Yoon, X. Jin, Customization of on-site assembly services by integrating the internet of things and BIM technologies in modular integrated construction, Autom. ConStruct. 126 (2021) 103663, https://doi.org/10.1016/j.autcon.2021.103663.
[72] L. Zhao, Z. Liu, J. Mbachu, Development of intelligent prefabs using IoT technology to improve the performance of prefabricated construction projects, Sensors 19 (19) (2019) 4131, https://doi.org/10.3390/s19194131.
[73] D. Lee, S.H. Lee, N. Masoud, M.S. Krishnan, V.C. Li, Integrated digital twin and blockchain framework to support accountable information sharing in construction projects, Autom. ConStruct. 127 (2021) 103688, https://doi.org/10.1016/j.autcon.2021.103688.
[74] Y. Zhao, C. Cao, Z. Liu, E. Mu, Intelligent control method of hoisting prefabricated components based on internet-of-things, Sensors 21 (3) (2021) 980, https://doi.org/10.3390/s21030980.
[75] Y. Zhong, Z. Qin, Positioning of prefabricated building components based on BIM and laser image scanning technology in the environment of internet of things, Comput. Intell. Neurosci. 2022 (2022), https://doi.org/10.1155/2022/5976935.
[76] H. Tran, T.N. Nguyen, P. Christopher, D.K. Bui, K. Khoshelham, T.D. Ngo, A digital twin approach for geometric quality assessment of as-built prefabricated façades, J. Build. Eng. 41 (2021) 102377, https://doi.org/10.1016/j.jobe.2021.102377.
[77] Z. Liu, A. Li, Z. Sun, G. Shi, X. Meng, Digital twin-based risk control during prefabricated building hoisting operations, Sensors 22 (7) (2022) 2522, https://doi.org/10.3390/s22072522.
[78] C.Z. Li, Z. Chen, F. Xue, X.T. Kong, B. Xiao, X. Lai, Y. Zhao, A blockchain-and IoT-based smart product-service system for the sustainability of prefabricated housing construction, J. Clean. Prod. 286 (2021) 125391, https://doi.org/10.1016/j.jclepro.2020.125391.
[79] R. Zhao, Z. Chen, F. Xue, A blockchain 3.0 paradigm for digital twins in construction project management, Autom. ConStruct. 145 (2023) 104645, https://doi.org/10.1016/j.autcon.2022.104645.